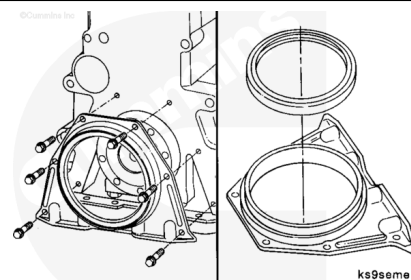


Trainings ▾

General Information

Note : This procedure is for front gear train engines only.

For front gear train engines, a rear crankshaft seal carrier is mounted to the rear of the cylinder block to house the rear crankshaft seal. For rear gear train engines, the rear crankshaft seal is installed in the flywheel housing.

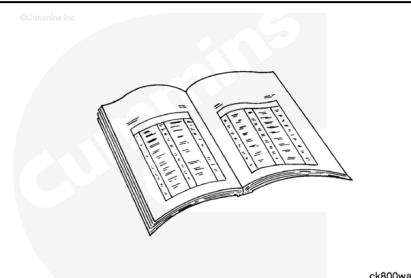


Preparatory Steps

⚠ WARNING ⚠

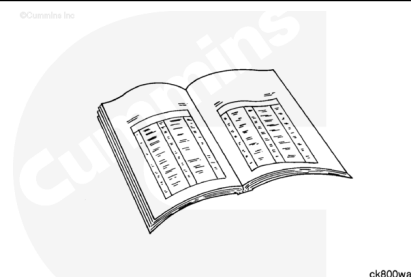
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. Refer to the OEM service manual.



Note : Use a container that can hold at least 26 liters [27 qt] of lubricating oil.

- If equipped with a wet flywheel housing, drain the oil from the flywheel housing by removing the plug in the bottom of the flywheel housing
- Remove the starting motor. Refer to Procedure 013-020 in Section 13 (/qs3/pubsys2/xml/en/procedures/40/40-013-020-tr.html)
- Remove the transmission and all related components (if equipped). Refer to the OEM instructions



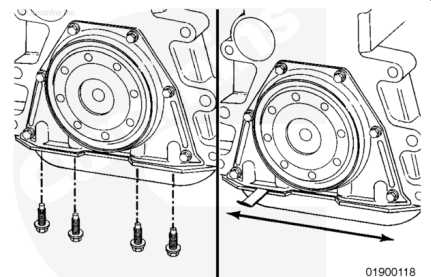
- Remove the flywheel/flexplate assembly. Refer to Procedure 016-005 in Section 16 (</qs3/pubsys2/xml/en/procedures/40/40-016-005-tr.html>) or Refer to Procedure 016-004 in Section 16 (</qs3/pubsys2/xml/en/procedures/40/40-016-004-tr.html>)
- Remove any OEM attached components (mufflers, shift mechanisms, air filters, etc.) to the flywheel housing. Refer to the OEM service manual
- Remove the flywheel housing. Refer to Procedure 016-006 in Section 16 (</qs3/pubsys2/xml/en/procedures/40/40-016-006-tr.html>).

Remove

Remove the four oil pan mounting capscrews that secure the oil pan to the rear cover.

⚠ CAUTION ⚠

Use extreme care when releasing the oil pan gasket from the rear cover to prevent damage to the gasket.

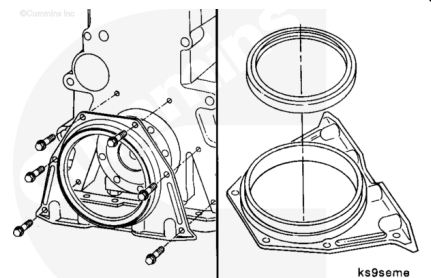


Insert the feeler gauge or shim stock between the rear cover and the oil pan gasket. Move the feeler gauge back and forth to release the gasket from the rear cover.

Remove the capscrews from the rear cover.

Remove the cover from the crankshaft flange.

Support the rear seal carrier on a flat work surface with wooden blocks. Use a suitable punch and hammer to drive the old seal out of the rear seal carrier.



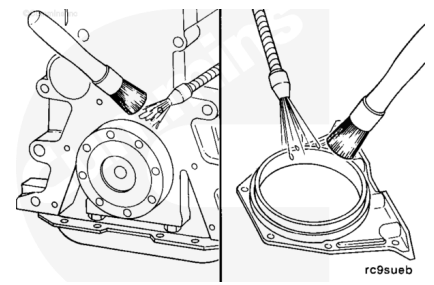
Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.

⚠ WARNING ⚠

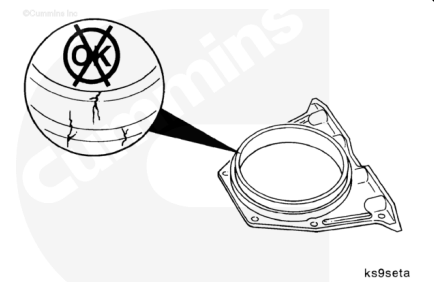
Compressed air used for cleaning should not exceed 207 kPa [30 psi]. Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.



Clean the gasket surface of the cylinder block and rear seal carrier.

Dry these areas with compressed air.

Inspect the rear seal carrier for cracks or other damage. Replace the rear seal carrier if any damage is found.



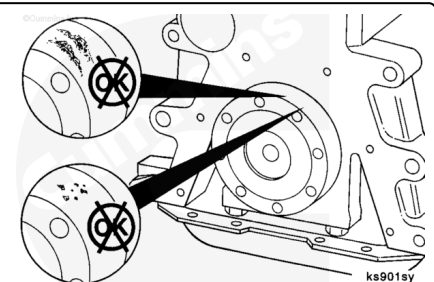
Use a fine crocus cloth to remove any rust or other deposits from the crankshaft flange.

Use a clean cloth and solvent to clean the crankshaft flange.

Inspect the crankshaft flange for nicks or burrs. Use an abrasive pad, Part Number 3823258 or equivalent, to remove any minor damage.

Note : If the crankshaft has excessive wear, a service wear sleeve is available.

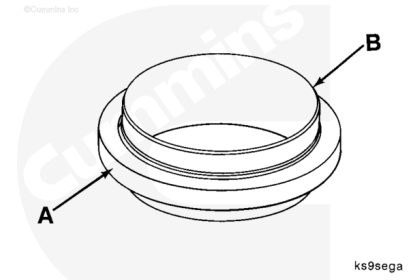
For engines equipped with a lip style rear crankshaft seal, inspect the crankshaft seal contact area for a wear groove. If the groove is deeper than 0.25 mm [0.010 in], a wear sleeve and oversize seal **must** be used. Refer to Procedure 001-067 in Section 1 (</qs3/pubsys2/xml/en/procedures/40/40-001-067.html>).



Install

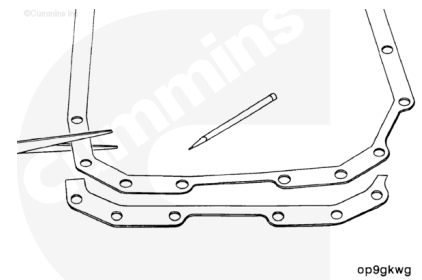
Note : For engines equipped with a wet flywheel housing, make sure to replace the rear crankshaft seal with the correct rear seal. Rear crankshaft seals for a wet and dry flywheel housings may not be the same.

The new rear crankshaft seal (A) will come installed on a seal pilot (B). The seal **must** be left on the seal pilot while installing the seal onto the nose of the crankshaft. This will keep the lips of the seal from being damaged during installation.



Service Tip: If the oil pan gasket was damaged during removal of the rear seal carrier, remove the damaged gasket and:

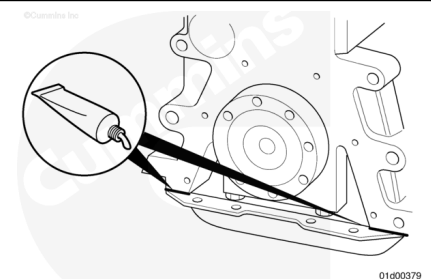
1. If previously equipped with a paper oil pan gasket, use the old gasket as a pattern and cut a section of a new gasket to the same size. Use a light coat of sealant, Part Number 3164067, to hold the gasket in place
2. If previously equipped with a sealant only oil pan gasket, apply a bead of sealant, Part Number 3164070, to the oil pan flange
3. If previously equipped with a suspended oil pan, the oil pan must be removed and a new gasket installed. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html>).



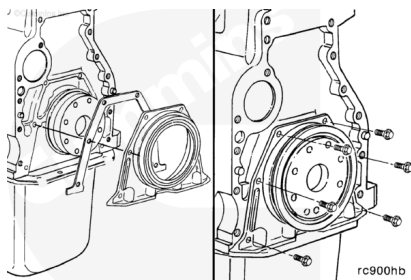
Note : The rear seal carrier **must** be installed within 10 minutes of applying the sealant.

Note : It may be necessary to trim the rear seal carrier gasket so that it is even with the oil pan mounting surface. Test fit the gasket prior to installing.

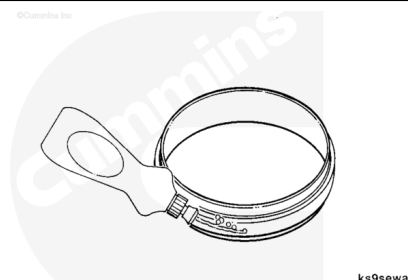
Apply a thin bead of sealant, Part Number 3164067, at the intersecting joint of the oil pan and cylinder block prior to installing the rear crankshaft seal carrier. The rear crankshaft seal carrier **must** be installed within 10 minutes of applying the sealant.



Install the rear crankshaft seal carrier and gasket. Apply Loctite™ 262 or 271 to the mounting capscrews and loosely tighten the rear seal carrier to the block.



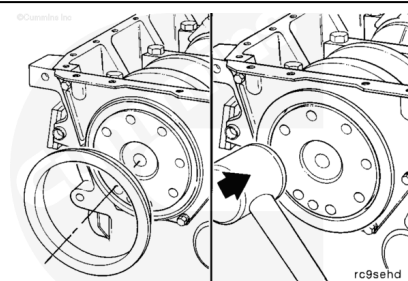
To aid in installation, the lubricating oil seal requires the application of a mild soap on the outside diameter of the seal case.



Place the new rear crankshaft seal, with the seal pilot, over the crankshaft nose and slide it by hand toward the flywheel housing.

Remove the seal pilot.

Use the disposable seal driver for front gear train engines to drive the rear crankshaft seal into the rear crankshaft seal carrier.



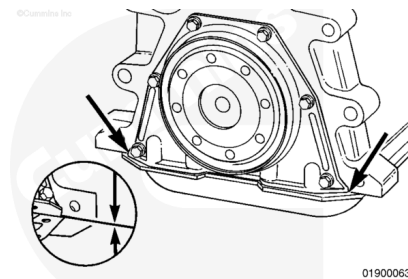
Note : Service tool, Part Number 3824078, used to install the rear crankshaft seal/wear sleeve assembly, can also be used to install the rear crankshaft seal.

Use a plastic hammer to drive the seal into the housing until the alignment tool stops against the housing.

Hit the tool at the 12, 3, 6 and 9 o'clock positions to drive the seal evenly and to prevent bending the seal carrier. Hit the seal driver until contact is made with the rear crankshaft seal carrier.

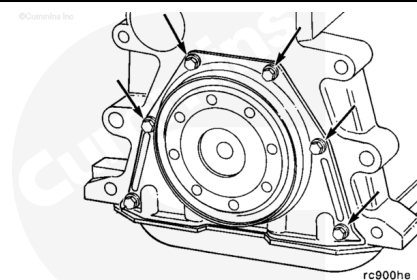
⚠ CAUTION ⚠

Do not push or force the cover in any direction. This may cause an irregular seal lip position after seal installation. An engine oil leak will result.



Align the rear cover even with both sides of the oil pan rail on the cylinder block.

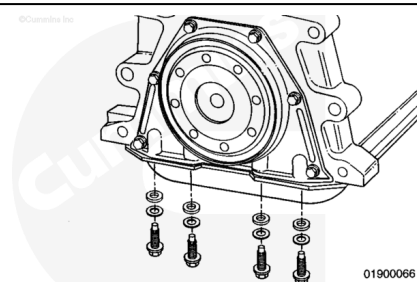
Tighten the rear cover capscrews. The chart below shows the proper torque value when using either a 9.8 or 10.9 grade capscrew. The grade is embossed on the top of the capscrew.



Rear Seal Carrier				
Bolt Number	Bolt Size	Bolt Class	Torque	Dri-Loc™
3913638	M-6	9.8	10 N•m [89 in-lb]	Yes
3991306	M-6	10.9	13 N•m [115 in-lb]	Yes

Install the four rear oil pan mounting capscrews into the pan.

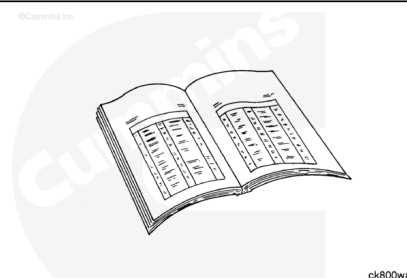
Torque Value: 24 n•m [212 in-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the flywheel/flexplate assembly. Refer to Procedure 016-005 (/qs3/pubsys2/xml/en/procedures/40/40-016-005-tr.html) in Section 16 or Refer to Procedure 016-004 in Section 16 (/qs3/pubsys2/xml/en/procedures/40/40-016-004-tr.html)
- Install the starting motor. Refer to Procedure 013-020 in Section 13 (/qs3/pubsys2/xml/en/procedures/40/40-013-020-tr.html)
- Install the transmission and all related components (if equipped). Refer to the OEM service manual

- If equipped with a wet flywheel housing, fill the flywheel housing with oil. Refer to OEM service manual
- If previously removed, attach any OEM attached components (mufflers, shift mechanisms, air filters, etc.) to the flywheel housing. Refer to the OEM service manual.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries. Refer to the OEM service manual.
- Operate engine and check for leaks.

